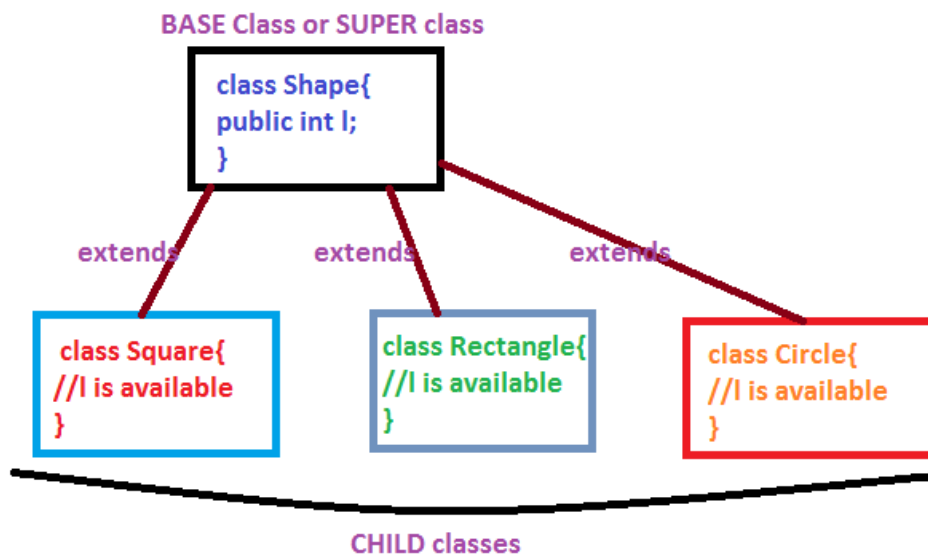


OOPS Concept

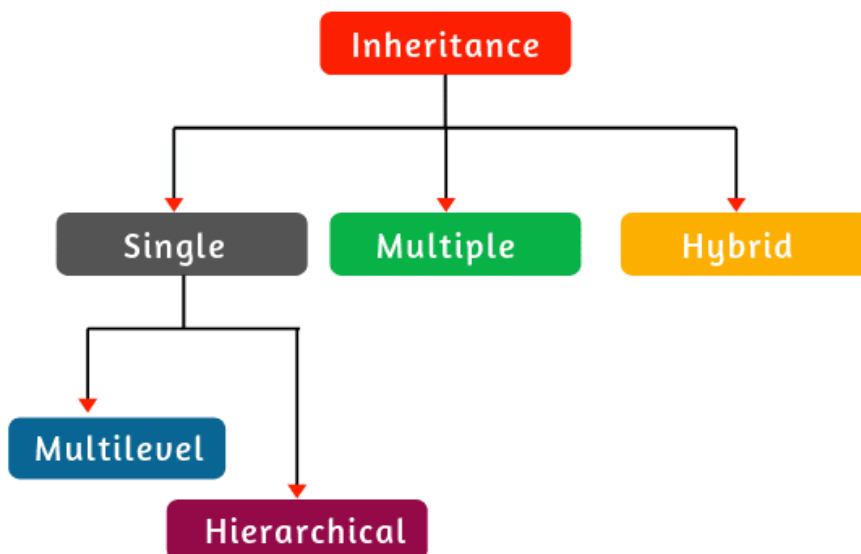
Inheritance in Java :-

- It is the process where one class acquires fields & methods of another.
- The child class can use fields and methods of the parent class.
- extends keyword is used to inherit a class.
- Inheritance promotes code reusability.
- Java supports single inheritance (one class can inherit only one class directly), but supports multilevel inheritance and hierarchical inheritance.



Types of Inheritance :-

- Java supports the following main types of inheritance:



1. Single Inheritance –

- One class inherits another class.
- Achieved using extends keyword.
- Child class gets parent's properties + methods.
- Promotes code reusability.

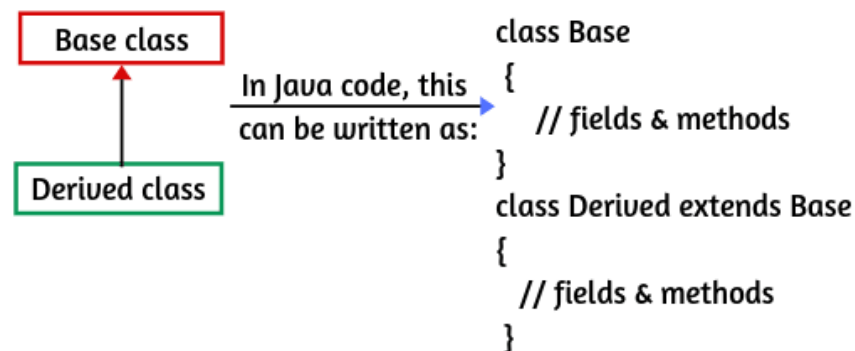


Fig: Single level Inheritance

Example –

```
package oops;
class parent {
    void first()
    {
        System.out.println("Parent Property");
    }
}

public class single_inheritance extends parent{
    void second()
    {
        System.out.println("Child Property");
    }

    public static void main(String[] args) {
        single_inheritance c=new single_inheritance();
        c.second();
        c.first();
    }
}
```

Output –

Child Property

Parent Property

2. Multilevel Inheritance –

- In multilevel inheritance, a class inherits from another class, which itself inherits from another class.
- Forms a chain of inheritance.
- Achieved using extends keyword.

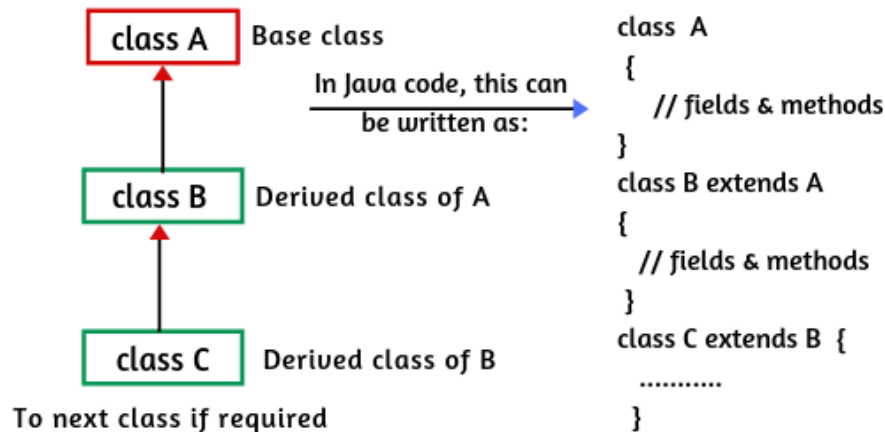


Fig: Multilevel Inheritance in Java

Example –

```
package oops;

class multilevel_grandparent{
    void mgp()
    {
        System.out.println("Grand Parent Class");
    }
}
class multilevel_parent extends multilevel_grandparent{
    void mp()
    {
        System.out.println("Parent Class");
    }
}
public class multilevel_inheritance extends multilevel_parent{
    void mc()
    {
        System.out.println("Child Class");
    }
    public static void main(String[] args) {
        multilevel_inheritance c=new multilevel_inheritance();
        c.mc();
        c.mp();
        c.mgp();
    }
}
```

Output –

Child Class

Parent Class

Grand Parent Class